

Reconstruction of Nonlinear Susceptibility Tensors Using a Symmetry-Aware Neural Network

Abstract

This project develops a physics-informed deep learning framework for reconstructing nonlinear optical susceptibility tensor elements from rotational anisotropy second harmonic generation, or RA-SHG, intensity patterns. RA-SHG is sensitive to crystal symmetry and is widely useful for studying nonlinear optical response, structural anisotropy, and phase behavior in condensed matter systems. However, extracting the nonlinear susceptibility tensor from measured intensity data is an inverse problem: the measured signal is proportional to the squared magnitude of the nonlinear polarization, so phase and sign information are not directly available.

The project addresses this problem by generating synthetic RA-SHG intensity curves for two point groups, D_{3h} and C_s , and training a one-dimensional convolutional neural network to reconstruct four tensor parameters:

$$[d_{11}, d_{22}, d_{12}, d_{26}]. \quad (1)$$

A central feature of the model is the use of crystallographic symmetry constraints. The network is conditioned on the point group and applies a hard symmetry mask to the output, ensuring that forbidden tensor elements are exactly zero. This implements Neumann’s principle directly in the architecture rather than treating symmetry as a soft penalty.

The trained model achieves a final test mean squared error of approximately

$$\text{Test MSE} = 0.0130, \quad (2)$$

on a separately generated synthetic test set. The results show that the model successfully learns symmetry-constrained tensor reconstruction from noisy RA-SHG patterns, while strictly enforcing physically allowed tensor structures.

1 Introduction

Second harmonic generation, or SHG, is a second-order nonlinear optical process in which two incident photons at frequency ω generate one photon at frequency 2ω . In a noncentrosymmetric medium, the induced nonlinear polarization is related to the electric field by

$$P_i(2\omega) = \sum_{jk} \chi_{ijk}^{(2)} E_j(\omega) E_k(\omega). \quad (3)$$

Using contracted Voigt notation, this relation can be written as

$$P_i = \sum_{J=1}^6 d_{iJ} E_J, \quad (4)$$

where the Voigt vector is

$$\mathbf{E}_{\text{Voigt}} = \begin{bmatrix} E_x^2 \\ E_y^2 \\ E_z^2 \\ 2E_y E_z \\ 2E_x E_z \\ 2E_x E_y \end{bmatrix}. \quad (5)$$

In rotational anisotropy SHG, the incident polarization is rotated through an angle ϕ , and the SHG intensity is recorded as a function of angle. The angular dependence of the intensity contains information about the nonlinear susceptibility tensor and the crystal symmetry.

However, the inverse problem is nontrivial because the measured intensity is

$$I(\phi) = |P_{\parallel}(\phi)|^2, \quad (6)$$

rather than the complex nonlinear polarization itself. Since the phase information is lost, different susceptibility tensors may produce similar intensity curves. This makes tensor reconstruction a phase-retrieval-type problem.

The project uses a neural network to solve this inverse problem, but constrains the model using known crystallographic symmetry. This is physically important because the susceptibility tensor cannot take arbitrary values: the point group of the crystal determines which tensor elements are allowed and which must vanish.

2 Project Objective

The goal of the project is to reconstruct nonlinear susceptibility tensor elements from RA-SHG intensity curves using a physics-informed neural network.

The main objectives are:

1. Generate synthetic RA-SHG data using a forward nonlinear optical model.
2. Simulate two point groups: D_{3h} and C_s .
3. Train a one-dimensional convolutional neural network to infer tensor parameters from angular intensity data.
4. Include point-group information as an input to the model.
5. Enforce crystallographic symmetry through a hard output mask.
6. Evaluate the model on an independent synthetic test set.

The model predicts four tensor parameters:

$$[d_{11}, d_{22}, d_{12}, d_{26}]. \quad (7)$$

The predicted elements are not all allowed for every point group. The allowed structure depends on symmetry.

3 Forward RA-SHG Model

3.1 Electric Field Geometry

The simulation assumes normal incidence and an in-plane linearly polarized incident field. At rotation angle ϕ , the field components are

$$E_x = \cos \phi, \quad E_y = \sin \phi, \quad E_z = 0. \quad (8)$$

Therefore, the Voigt vector becomes

$$\mathbf{E}_{\text{Voigt}}(\phi) = \begin{bmatrix} \cos^2 \phi \\ \sin^2 \phi \\ 0 \\ 0 \\ 0 \\ 2 \cos \phi \sin \phi \end{bmatrix}. \quad (9)$$

The nonlinear polarization is calculated using

$$\mathbf{P}(\phi) = d\mathbf{E}_{\text{Voigt}}(\phi), \quad (10)$$

where d is the 3×6 nonlinear susceptibility tensor in contracted notation.

3.2 Parallel Polarization Detection

The notebook uses a parallel polarization detection geometry. The detected polarization component is

$$P_{\parallel}(\phi) = P_x(\phi) \cos \phi + P_y(\phi) \sin \phi. \quad (11)$$

The SHG intensity is then calculated as

$$I(\phi) = P_{\parallel}^2(\phi). \quad (12)$$

Gaussian noise is added to the simulated intensity curve to mimic experimental detector noise or laser fluctuations. Negative noisy intensities are clipped to zero because physical intensity cannot be negative.

4 Point-Group Symmetry and Tensor Structure

A key physical principle used in the project is that the susceptibility tensor must obey the symmetry of the crystal. This is an application of Neumann's principle: the physical properties of a crystal must remain invariant under the symmetry operations of the crystal's point group.

The project studies two point groups: D_{3h} and C_s .

4.1 D_{3h} Symmetry

For the D_{3h} point group, the model uses one independent in-plane tensor parameter, d_{22} . The implemented tensor structure is

$$d_{16} = -d_{22}, \quad d_{21} = -d_{22}, \quad d_{22} = d_{22}. \quad (13)$$

In matrix form, the relevant nonzero components are placed as

$$d_{0,5} = -d_{22}, \quad d_{1,0} = -d_{22}, \quad d_{1,1} = d_{22}. \quad (14)$$

For the four predicted parameters,

$$[d_{11}, d_{22}, d_{12}, d_{26}], \quad (15)$$

the D_{3h} target structure is

$$[0, d_{22}, 0, 0]. \quad (16)$$

Thus, only d_{22} is allowed to be nonzero.

4.2 C_s Symmetry

For the C_s point group, the project allows three independent in-plane tensor elements:

$$d_{11}, \quad d_{12}, \quad d_{26}. \quad (17)$$

The implemented tensor structure is

$$d_{0,0} = d_{11}, \quad d_{0,1} = d_{12}, \quad d_{1,5} = d_{26}. \quad (18)$$

For the predicted vector

$$[d_{11}, d_{22}, d_{12}, d_{26}], \quad (19)$$

the C_s target structure is

$$[d_{11}, 0, d_{12}, d_{26}]. \quad (20)$$

Thus, d_{22} is forbidden for C_s in the implemented model.

5 Dataset Generation

The project generates a synthetic RA-SHG dataset using the forward model described above.

The angular range is sampled from 0° to 360° using 100 equally spaced points. Each data sample consists of:

- the point group label,
- the tensor labels $[d_{11}, d_{22}, d_{12}, d_{26}]$,
- 100 intensity values $I_0, I_1, \dots, I_{99}$.

For each point group, the allowed tensor parameters are randomly sampled from a uniform distribution between 0 and 1.

For D_{3h} ,

$$d_{22} \sim U(0, 1), \quad (21)$$

while

$$d_{11} = d_{12} = d_{26} = 0. \quad (22)$$

For C_s ,

$$d_{11}, d_{12}, d_{26} \sim U(0, 1), \quad (23)$$

while

$$d_{22} = 0. \quad (24)$$

Noise is randomly chosen between 1% and 10% for each generated pattern:

$$\text{noise level} \sim U(0.01, 0.10). \quad (25)$$

The training dataset contains

$$10,000 \quad (26)$$

samples for D_{3h} and

$$10,000 \quad (27)$$

samples for C_s , giving a total of

$$20,000 \quad (28)$$

synthetic training samples.

A separate test dataset is generated with

$$2,000 \quad (29)$$

samples per point group, for a total of

$$4,000 \quad (30)$$

test samples.

6 Neural Network Architecture

The neural network used in the project is called **AdvancedPhysicsSHGNet**. It is a one-dimensional convolutional neural network designed to process angular RA-SHG intensity patterns.

6.1 Input Representation

The network receives two inputs:

1. A 100-point RA-SHG intensity curve.
2. A one-hot encoded point-group vector.

The point-group encoding is

$$D_{3h} = [1, 0], \quad C_s = [0, 1]. \quad (31)$$

Including the point group as an input is important because the same intensity reconstruction task depends on the symmetry class. The network is not merely learning a generic mapping from intensity to tensor; it is learning a symmetry-conditioned inverse map.

6.2 Convolutional Layers

The model uses two one-dimensional convolutional layers.

The first convolutional block is

$$\text{Conv1D}(1 \rightarrow 32, k = 7), \quad (32)$$

followed by batch normalization, ReLU activation, and max pooling. The pooling reduces the input length from 100 to 50.

The second convolutional block is

$$\text{Conv1D}(32 \rightarrow 64, k = 5), \quad (33)$$

again followed by batch normalization, ReLU activation, and max pooling. The pooling reduces the length from 50 to 25.

A physically meaningful design choice is the use of circular padding. Since RA-SHG intensity curves are periodic over 360° , the data point near 0° is physically adjacent to the data point near 360° . Circular padding respects this periodic boundary condition.

6.3 Fully Connected Layers

After convolution and pooling, the feature map has size

$$64 \times 25 = 1600. \quad (34)$$

This 1600-dimensional feature vector is concatenated with the two-dimensional point-group vector, giving a total input size of

$$1600 + 2 = 1602 \quad (35)$$

for the fully connected block.

The dense layers are

$$1602 \rightarrow 256 \rightarrow 4. \quad (36)$$

The final four outputs correspond to

$$[d_{11}, d_{22}, d_{12}, d_{26}]. \quad (37)$$

A dropout layer with dropout probability 0.1 is included after the first fully connected layer.

7 Hard Symmetry Mask

The most important physics-informed component of the neural network is the hard symmetry mask. The mask matrix is

$$M = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}. \quad (38)$$

The first row corresponds to D_{3h} , and the second row corresponds to C_s . The columns correspond to

$$[d_{11}, d_{22}, d_{12}, d_{26}]. \quad (39)$$

Given a point-group one-hot vector $\mathbf{g}$, the model computes a dynamic mask

$$\mathbf{m} = \mathbf{g}M. \quad (40)$$

The raw tensor prediction from the neural network is then multiplied element-wise by this mask:

$$\hat{\mathbf{d}}_{\text{final}} = \hat{\mathbf{d}}_{\text{raw}} \odot \mathbf{m}. \quad (41)$$

This means the model output is forced to obey symmetry exactly.

For D_{3h} ,

$$\mathbf{m}_{D_{3h}} = [0, 1, 0, 0], \quad (42)$$

so

$$\hat{\mathbf{d}}_{D_{3h}} = [0, \hat{d}_{22}, 0, 0]. \quad (43)$$

For C_s ,

$$\mathbf{m}_{C_s} = [1, 0, 1, 1], \quad (44)$$

so

$$\hat{\mathbf{d}}_{C_s} = [\hat{d}_{11}, 0, \hat{d}_{12}, \hat{d}_{26}]. \quad (45)$$

This is stronger than simply adding a penalty term to the loss function. The forbidden components are mathematically forced to zero by construction.

8 Training Procedure

The synthetic dataset is split into training and validation sets using an 80/20 split. Since the full training dataset contains 20,000 samples, this gives approximately

$$16,000 \tag{46}$$

training samples and

$$4,000 \tag{47}$$

validation samples.

The model is trained using the Adam optimizer with learning rate

$$\eta = 0.005. \tag{48}$$

The batch size is

$$128. \tag{49}$$

The maximum number of epochs is 50, but early stopping is used with patience 5. The loss function is mean squared error:

$$\text{MSE} = \frac{1}{N} \sum_i (\hat{d}_i - d_i)^2. \tag{50}$$

Because the model output is already symmetry-masked, standard MSE is sufficient to compare the final physical prediction against the target tensor.

The training log shows rapid convergence:

$$\text{Epoch 1: Train Loss} = 0.4701, \quad \text{Validation Loss} = 0.0144. \tag{51}$$

By epoch 5,

$$\text{Train Loss} = 0.0169, \quad \text{Validation Loss} = 0.0117. \tag{52}$$

By epoch 10,

$$\text{Train Loss} = 0.0157, \quad \text{Validation Loss} = 0.0119. \tag{53}$$

Early stopping is triggered at epoch 12, with the best validation loss

$$\boxed{\text{Best Validation Loss} = 0.0114.} \tag{54}$$

The best model weights are restored after early stopping.

9 Test Results

The trained model is evaluated on a separately generated test dataset containing 4,000 samples. The reported final test loss is

$$\boxed{\text{Final Test MSE} = 0.0130.} \tag{55}$$

This test loss is close to the best validation loss, suggesting that the model generalizes reasonably well to newly generated synthetic RA-SHG patterns.

Sample predictions for C_s -type test data are:

$$\text{Predicted} = [0.3714, 0, 0.5290, 0.4673], \quad (56)$$

while the corresponding target tensor is

$$\text{Actual} = [0.4201, 0, 0.0491, 0.7447]. \quad (57)$$

Another example is

$$\text{Predicted} = [0.9017, 0, 0.6919, 0.7586], \quad (58)$$

with target

$$\text{Actual} = [0.9371, 0, 0.3357, 0.9898]. \quad (59)$$

A third example is

$$\text{Predicted} = [0.4955, 0, 0.5067, 0.4289], \quad (60)$$

with target

$$\text{Actual} = [0.5567, 0, 0.8757, 0.2490]. \quad (61)$$

These examples show that the network correctly enforces the symmetry condition $d_{22} = 0$ for C_s . The allowed components are reconstructed with moderate accuracy; however, some individual tensor elements still show noticeable errors.

10 Performance Interpretation

The model performs well in two important ways.

First, it learns the symmetry structure exactly. Forbidden components are not merely small; they are exactly zero because of the hard mask. This is essential for physical reliability.

Second, the model reaches a relatively low test MSE of approximately 0.0130 on noisy synthetic data. The close agreement between validation and test losses indicates that the network is not simply memorizing the training set.

However, the sample predictions also show that the reconstruction of individual allowed tensor elements is not perfect. For example, in one C_s sample, the true value of d_{12} is approximately 0.0491, while the model predicts approximately 0.5290. This suggests that some tensor components may be partially degenerate in their contributions to the measured intensity pattern.

This behavior is physically understandable. Since the measured signal is an intensity,

$$I(\phi) = |P_{\parallel}(\phi)|^2, \quad (62)$$

the sign and phase information of the nonlinear polarization are lost. In addition, different combinations of tensor elements can produce similar angular intensity patterns, especially under noise. Therefore, the reconstruction problem is not a simple one-to-one regression problem.

The model is therefore best interpreted as a symmetry-constrained approximate inverse model: it can recover the physically allowed tensor structure and estimate tensor values, but some ambiguity remains due to the nature of intensity-only measurements.

11 Limitations

The main limitation is that the forward model is simplified. It assumes normal incidence and uses only an in-plane field with

$$E_z = 0. \quad (63)$$

As a result, out-of-plane tensor components are ignored. In real RA-SHG experiments, finite incidence angle, Fresnel coefficients, polarization geometry, surface contributions, and interference effects can be important.

A second limitation is that the model is trained entirely on synthetic data. Synthetic data are useful for controlled testing, but the model’s performance on real experimental data depends on how accurately the simulation captures the experimental setup.

A third limitation is the intrinsic ambiguity of intensity-only tensor reconstruction. Since the measured signal is proportional to the square of the nonlinear polarization, the inverse problem can contain degeneracies. This limits the ability of any regression model to uniquely recover all tensor elements from intensity alone.

A fourth limitation is that the implemented model considers only two point groups, D_{3h} and C_s . A more general tensor reconstruction framework would need to include additional point groups and a broader set of allowed tensor components.

12 Possible Improvements

Several improvements could strengthen the project.

First, the forward model could include a random angular offset:

$$I(\phi) \rightarrow I(\phi + \phi_0), \quad (64)$$

where ϕ_0 is sampled randomly. This would simulate experimental uncertainty in the crystal orientation.

Second, the model could be extended to additional polarization channels, such as parallel and crossed polarization geometries. Multiple polarization channels would provide more information and reduce tensor reconstruction ambiguity.

Third, the synthetic data could include more realistic experimental effects, such as finite incidence angle, Fresnel factors, polarization leakage, background offsets, and detector scaling.

Fourth, the neural network could be trained not only to predict tensor elements, but also to reconstruct the RA-SHG curve from the predicted tensor. This would create a hybrid inverse-forward consistency loss:

$$\mathcal{L} = \mathcal{L}_{\text{tensor}} + \lambda \mathcal{L}_{\text{intensity}}. \quad (65)$$

Such a loss would encourage predicted tensors to reproduce the observed intensity pattern.

Fifth, the model could report uncertainty estimates for each reconstructed tensor element. Since the inverse problem is partially degenerate, uncertainty quantification would be physically meaningful.

13 Conclusion

This project demonstrates a symmetry-aware neural network approach for reconstructing nonlinear susceptibility tensor elements from RA-SHG intensity data. The project combines a physical forward model, synthetic noisy data generation, a one-dimensional convolutional architecture, point-group conditioning, and a hard crystallographic symmetry mask.

The most important achievement is the exact enforcement of point-group symmetry. The model guarantees that forbidden tensor elements are zero for both D_{3h} and C_s , ensuring that predictions remain physically valid. The network achieves a final test MSE of approximately

$$\boxed{0.0130}, \tag{66}$$

showing that it can learn a useful inverse mapping from noisy RA-SHG patterns to susceptibility tensor parameters.

The project is a strong proof of concept for physics-informed machine learning in nonlinear optics. Its main limitations come from the simplified forward model, synthetic-only training data, and the intrinsic ambiguity of intensity-based phase retrieval. With more realistic experimental modeling, additional point groups, multiple polarization channels, and uncertainty quantification, this framework could become a useful tool for automated RA-SHG tensor analysis in condensed matter and nonlinear optical experiments.

References

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